

Ain Shams University

Faculty of Computer and Information Sciences

Big Data Analytics Project

Body Performance Analysis

Team Members

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Dataset: bodyPerformance.csv | Language: R
Academic Year 2025–2026

A. Project Description

This project applies the Big Data Analytics project lifecycle to a real-world body-performance dataset. Following the methodology studied in lectures and labs, the team carried out the full pipeline: data loading and cleaning, exploratory data analysis (EDA), statistical hypothesis testing, and supervised/unsupervised machine-learning modelling — all implemented in R.

The project was completed as a team effort by five members, each responsible for one phase of the pipeline:

Part	Member	Responsibility
Part 1	Menna	Data Loading, Preprocessing & EDA
Part 2	Moaz	Statistical Analysis & Hypothesis Testing
Part 3	Naira	Tree-Based ML Models (Decision Tree, Random Forest, XGBoost, Naive Bayes)
Part 4	Kareem	Logistic Regression (Multinomial) & KNN Model
Part 5	Mohamed	K-Means Clustering

The project deliverables include the cleaned dataset (.CSV), the full R script, and this documentation report covering all phases from data understanding through model evaluation.

B. Dataset and Variables Description

The dataset used is bodyPerformance.csv, a publicly available fitness and health dataset. It contains physical and physiological measurements for 13,393 individuals, with 12 attributes spanning anthropometric features, blood pressure readings, and fitness performance scores. The target variable is class, a performance rating assigned to each individual.

Dataset Overview

Property	Value
Number of Records	13,393 (after cleaning)
Number of Attributes	12 (plus 4 normalised columns added during preprocessing)
Target Variable	class (A, B, C, D)
File Format	CSV
Source	Body Performance Dataset (Kaggle)

Variable Descriptions

Variable	Type	Description
age	Numeric	Age of the individual in years (range: 20–64)
gender	Categorical	Gender of the individual: M (Male) or F (Female)
height_cm	Numeric	Height of the individual in centimetres
weight_kg	Numeric	Weight of the individual in kilograms
body fat_%	Numeric	Body fat percentage of the individual
diastolic	Numeric	Diastolic blood pressure reading (mmHg)
systolic	Numeric	Systolic blood pressure reading (mmHg)
gripForce	Numeric	Grip strength measured in kg (hand dynamometer)
sit and bend forward_cm	Numeric	Flexibility test: forward reach in centimetres
sit-ups counts	Numeric	Number of sit-ups performed in a single session
broad jump_cm	Numeric	Standing broad jump distance in centimetres
class	Categorical	Performance classification: A (best), B, C, D (lowest)

The class variable represents overall body-performance level, with A being the highest (best fitness) and D the lowest. It serves as the prediction target for all supervised classification models.

C. Problem Definition and Project Objectives

Problem Definition

Physical fitness assessment is increasingly important in healthcare, sports science, and wellness programmes. Manual grading of fitness performance is time-consuming and subjective. This project addresses the following research question:

"Can we accurately predict an individual's performance class (A/B/C/D) from measurable physical and physiological attributes?"

Secondary questions explored include: Does gender significantly affect grip strength? Does body fat percentage differ across performance classes? Which physical features are most predictive of performance class?

Project Objectives

1. Apply comprehensive data preprocessing to ensure data quality and suitability for machine learning.
2. Perform exploratory data analysis (EDA) using a variety of visualisations to extract insights about the dataset.
3. Test statistical hypotheses regarding gender-based grip differences and class-based body fat variation.
4. Train and evaluate multiple supervised classification algorithms: Decision Tree, Random Forest, XGBoost, Naive Bayes, Multinomial Logistic Regression, and KNN.
5. Apply unsupervised K-Means clustering to discover natural groupings in the data independent of class labels.
6. Compare model performance using accuracy, precision, recall, and F1-score, and identify the best-performing model.
7. Derive actionable findings about the physical attributes most strongly associated with high performance.

Data Science Objectives

- Type: Multi-class Supervised Classification (4 classes: A, B, C, D)
- Secondary: Unsupervised Clustering (K-Means, k=4)
- Evaluation Metrics: Accuracy, Precision, Recall, F1-Score, Confusion Matrix
- Train/Test Split: 70/30 (tree models) and 80/20 (regression & KNN), stratified

D. Data Visualization — Charts, Observations & Interpretations

A total of 16 visualisations were produced during the EDA phase to explore distributions, relationships, and class-specific patterns in the dataset.

Chart 1 — Combined Boxplots: Age, Weight, Body Fat, Grip Force

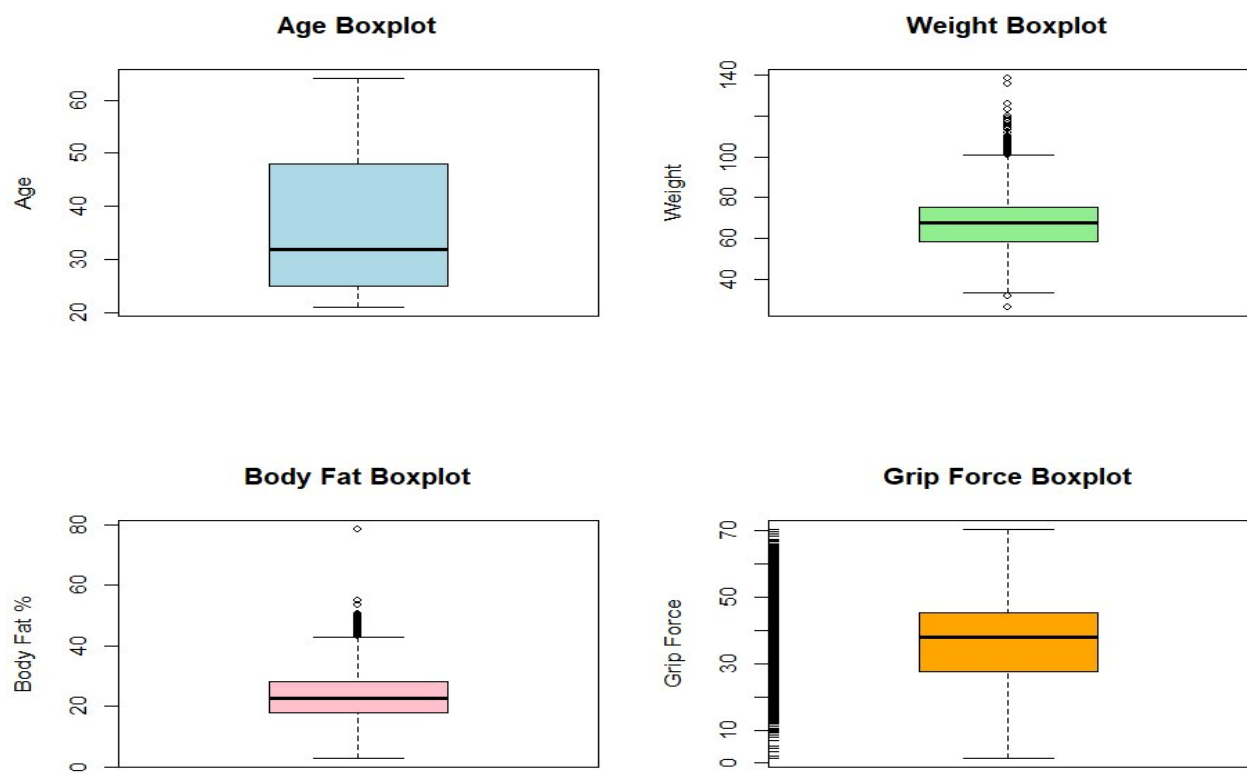


Figure 1: Boxplots for four key numerical variables

Observations:

- **Age:** The median age is approximately 32 years. The interquartile range spans roughly 25–47 years. No major outliers are present.
- **Weight:** The median weight is around 65 kg with a slight right skew, and several high-weight outliers are visible above 100 kg.
- **Body Fat:** Median body fat is approximately 21%, with a symmetric distribution and a few extreme outliers above 50%.

- Grip Force: Shows a wide spread (IQR ~25–47 kg) and a rug plot revealing a bimodal tendency — likely reflecting the male/female subpopulations.

Chart 2 — Class Distribution



Figure 2: Bar chart of performance class distribution

Observations:

- The dataset is perfectly balanced: each of the four classes (A, B, C, D) contains approximately 3,348 records.
- This balance is highly beneficial for machine learning, as it prevents class-imbalance bias and ensures that accuracy is a reliable performance metric without requiring resampling techniques.

Chart 3 — Gender Distribution

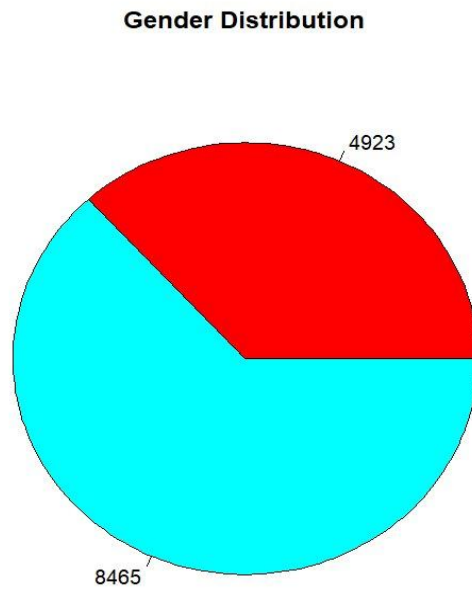


Figure 3: Pie chart of gender distribution

Observations:

- The dataset is imbalanced by gender: approximately 8,465 males (~63%) versus 4,923 females (~37%).
- This imbalance must be considered when interpreting gender-specific results. The larger male proportion could influence model predictions if gender is a strong feature.

Chart 4 — Age Distribution

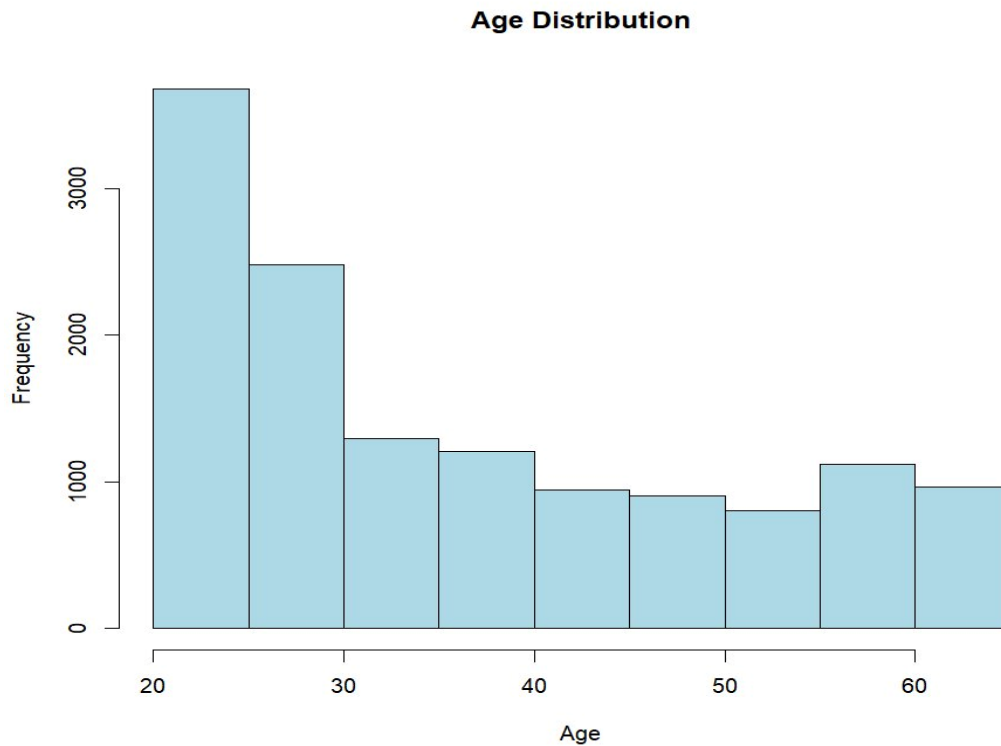


Figure 4: Histogram of age distribution

Observations:

- The distribution is right-skewed with the largest frequency in the 20–25 age group (over 3,500 individuals).
- Frequency declines steadily up to age 55, then shows a slight uptick in the 55–65 range — possibly a cohort effect or data collection pattern.
- The younger-skewed distribution means that the majority of performance observations come from younger adults.

Chart 5 — Weight Distribution

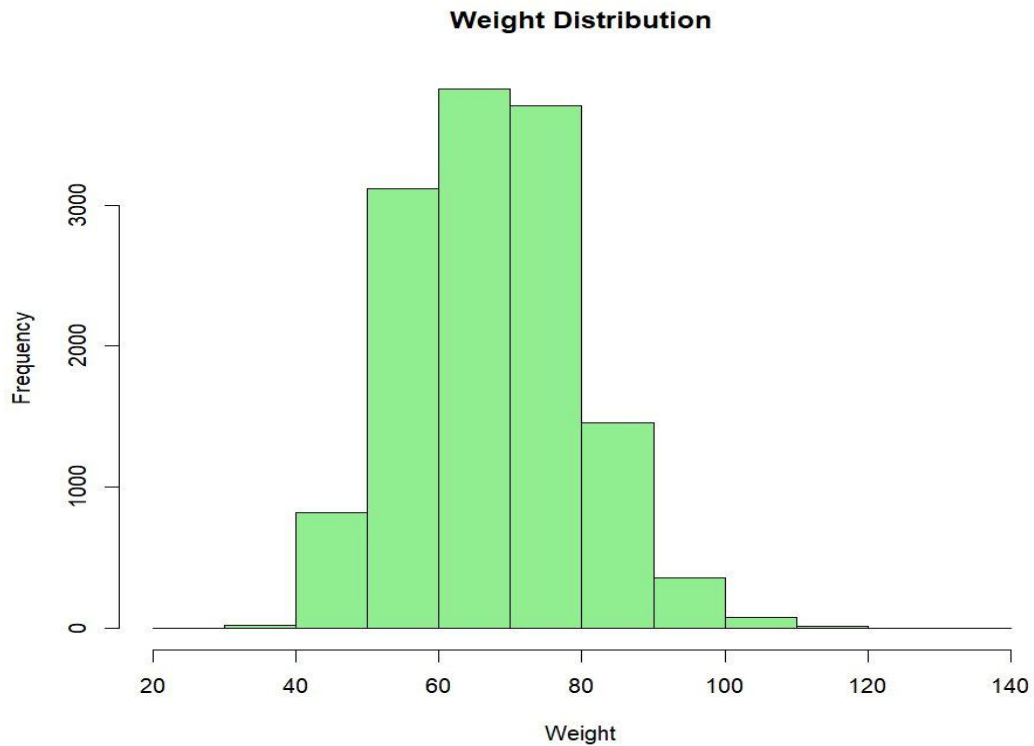


Figure 5: Histogram of weight distribution

Observations:

- Weight follows a roughly normal (bell-curve) distribution with a peak in the 60–80 kg range.
- A slight right skew is present, with a small proportion of individuals weighing over 90 kg.
- The distribution reflects a typical adult fitness population.

Chart 6 — Body Fat Distribution

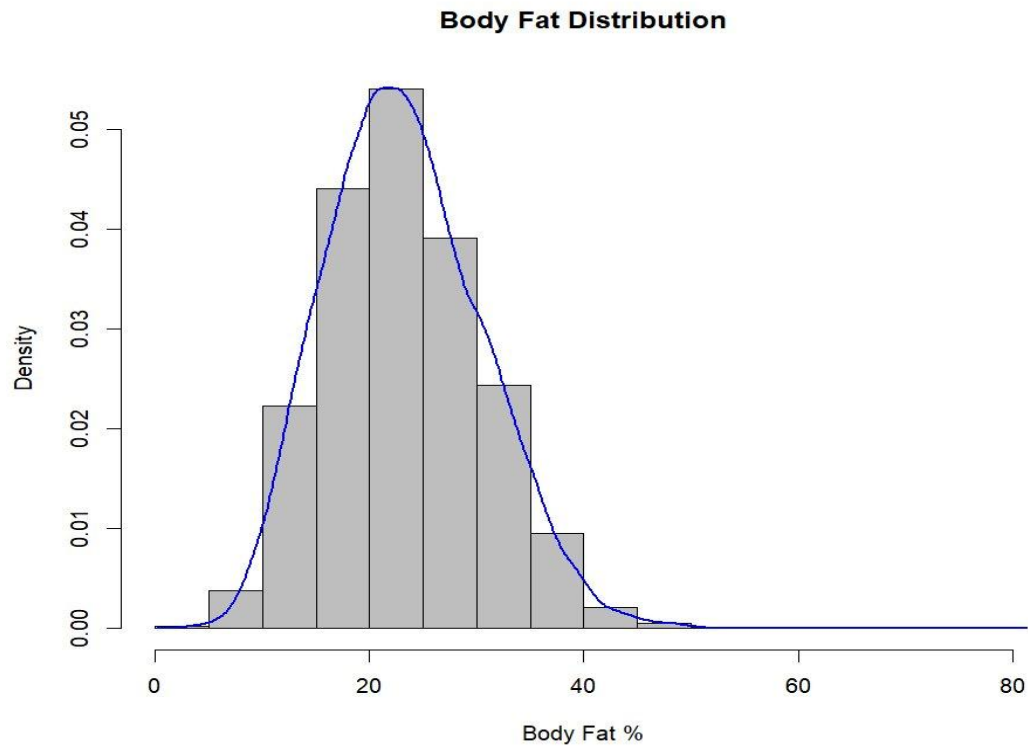


Figure 6: Histogram with KDE of body fat percentage

Observations:

- Body fat is approximately normally distributed with a peak density around 20–23%.
- The KDE curve closely follows the histogram shape, confirming near-normality.
- A slight right tail extends towards 50–80%, representing outliers who may have medical conditions or data entry errors.

Chart 7 — Grip Force Distribution

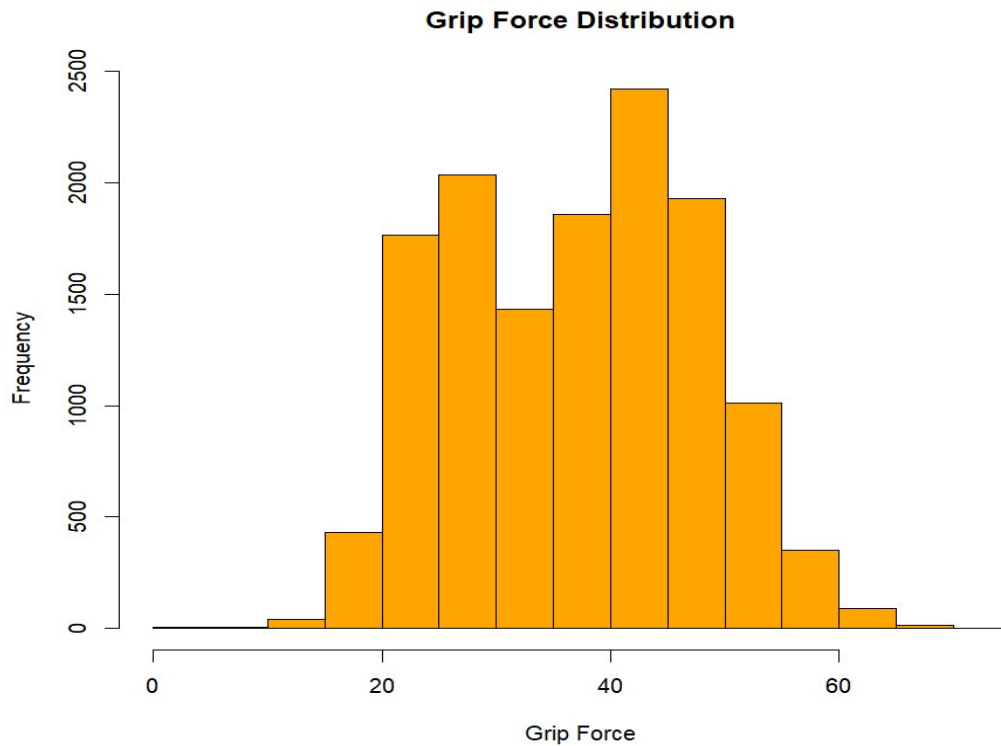


Figure 7: Histogram of grip force distribution

Observations:

- The distribution appears bimodal, with a dip between the 25–30 and 40–45 kg bins. This strongly suggests two underlying sub-populations: females (lower grip) and males (higher grip).
- The most frequent range is 40–45 kg, skewing the distribution towards higher values due to the larger male proportion in the dataset.

Chart 8 — Body Fat Percentage by Class

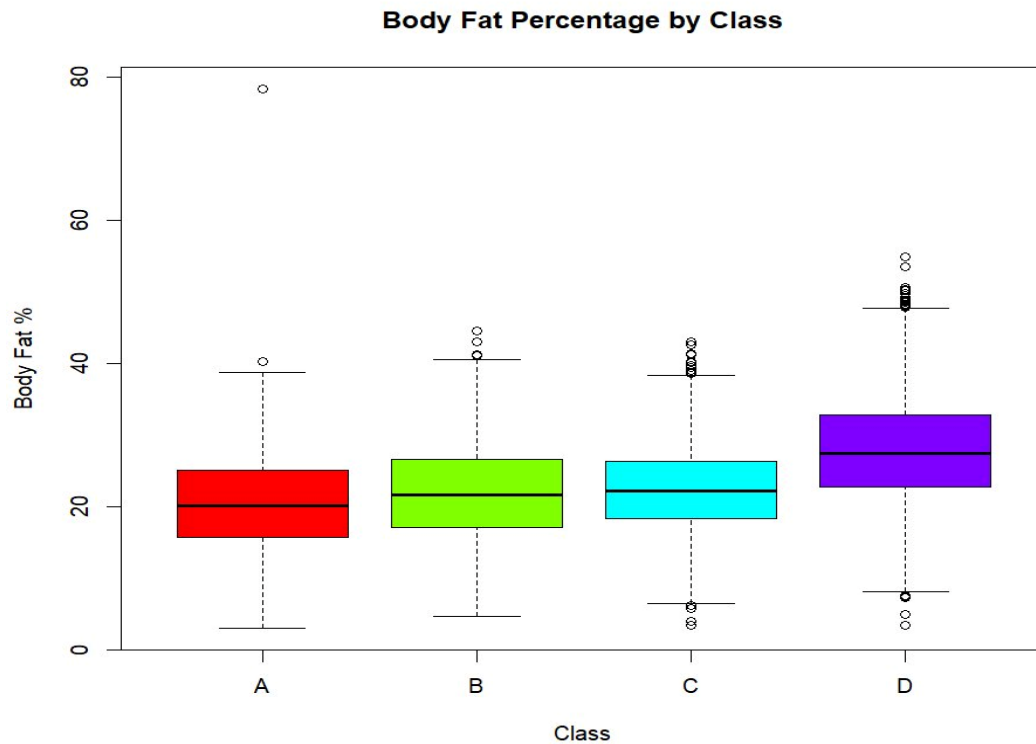


Figure 8: Boxplot of body fat % by performance class

Observations:

- Class D (lowest performance) has the highest median body fat (~27%), while Classes A, B, and C have similar medians (~20–22%).
- Classes A and B show tighter distributions, indicating more consistent body composition among higher performers.
- The outlier at ~78% in Class A is anomalous and may represent a data entry error or an exceptional individual.
- This chart supports the hypothesis that higher body fat is associated with lower performance class.

Chart 9 — Grip Force by Gender

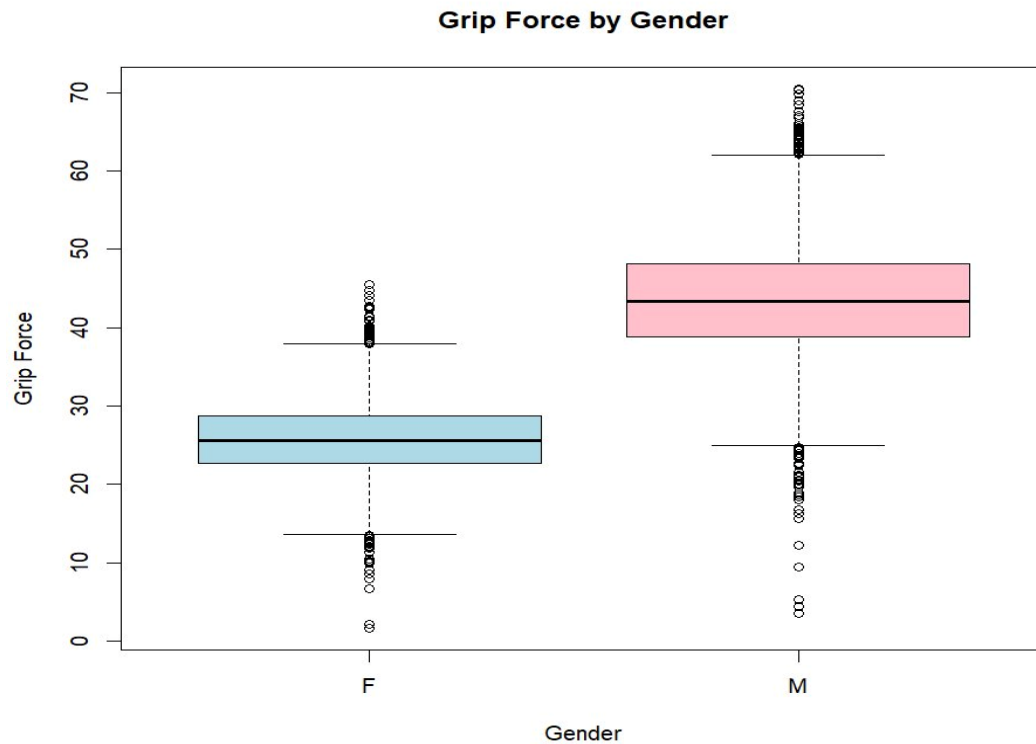


Figure 9: Boxplot of grip force by gender

Observations:

- Males have a substantially higher median grip force (~43 kg) compared to females (~25 kg) — nearly a 70% difference.
- The male IQR (38–48 kg) does not overlap with the female IQR (23–28 kg), visually confirming a statistically significant difference.
- This chart motivates and visually supports Hypothesis 1 (the independent t-test on grip force by gender), where the null hypothesis is decisively rejected.

Chart 10 — Sit-Ups Counts by Class

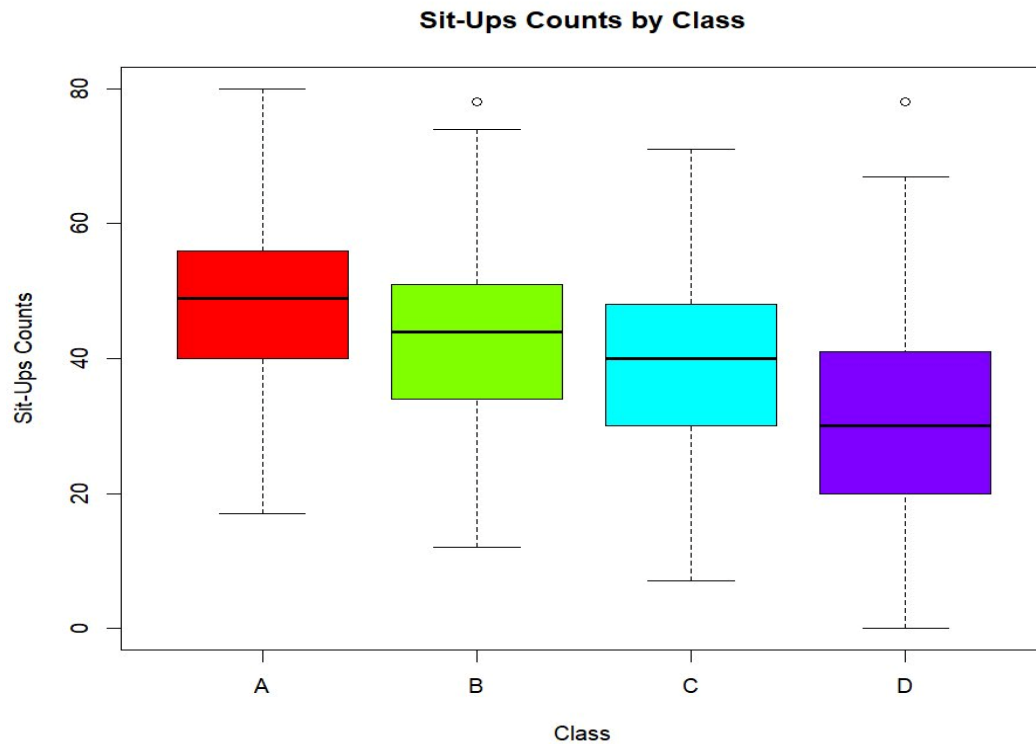


Figure 10: Boxplot of sit-ups counts by performance class

Observations:

- A clear decreasing trend in sit-up counts from Class A (median ~50) to Class D (median ~30) is visible.
- Class A has the tightest distribution at the high end, reflecting consistently strong performance.
- Class D shows the widest spread and lowest overall counts, suggesting high variability among the lowest performers.
- This confirms that sit-up capacity is a meaningful discriminator between performance classes.

Chart 11 — Age vs Sit-Ups Counts

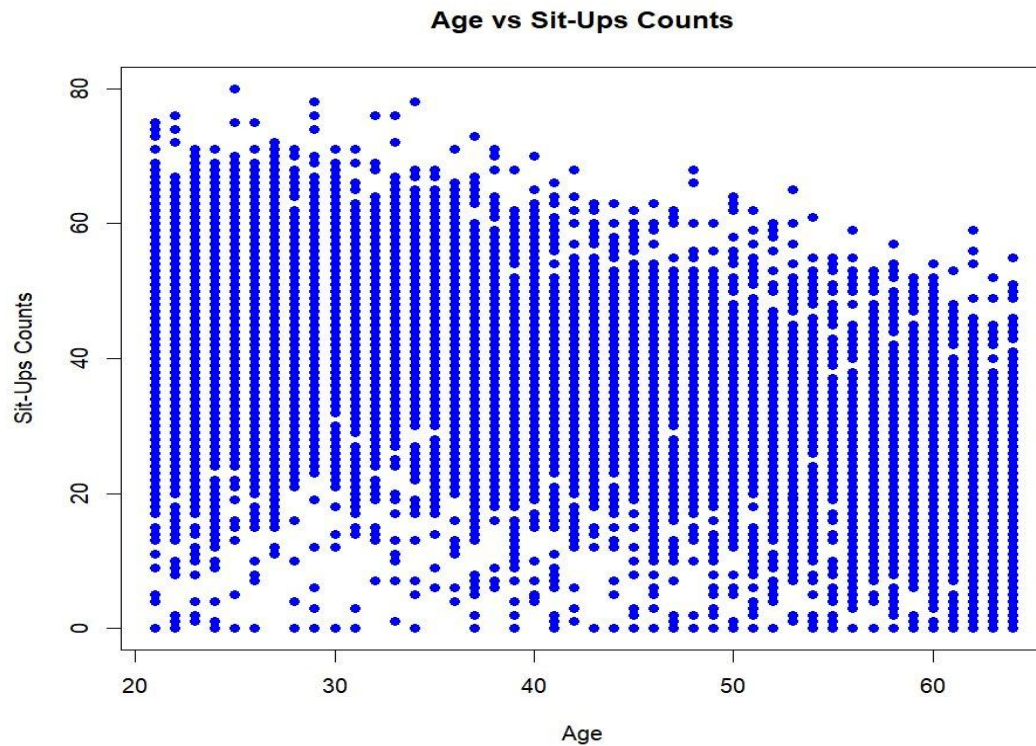


Figure 11: Scatter plot of age vs sit-ups counts

Observations:

- A clear negative trend is visible: sit-up performance declines as age increases. Younger individuals (20–30) achieve up to 80 sit-ups, while older individuals (50–64) are rarely seen above 60.
- The vertical striping is due to the integer nature of age values.
- This reinforces age as an important feature for performance classification, with physical endurance decreasing with advancing age.

Chart 12 — Grip Force vs Broad Jump

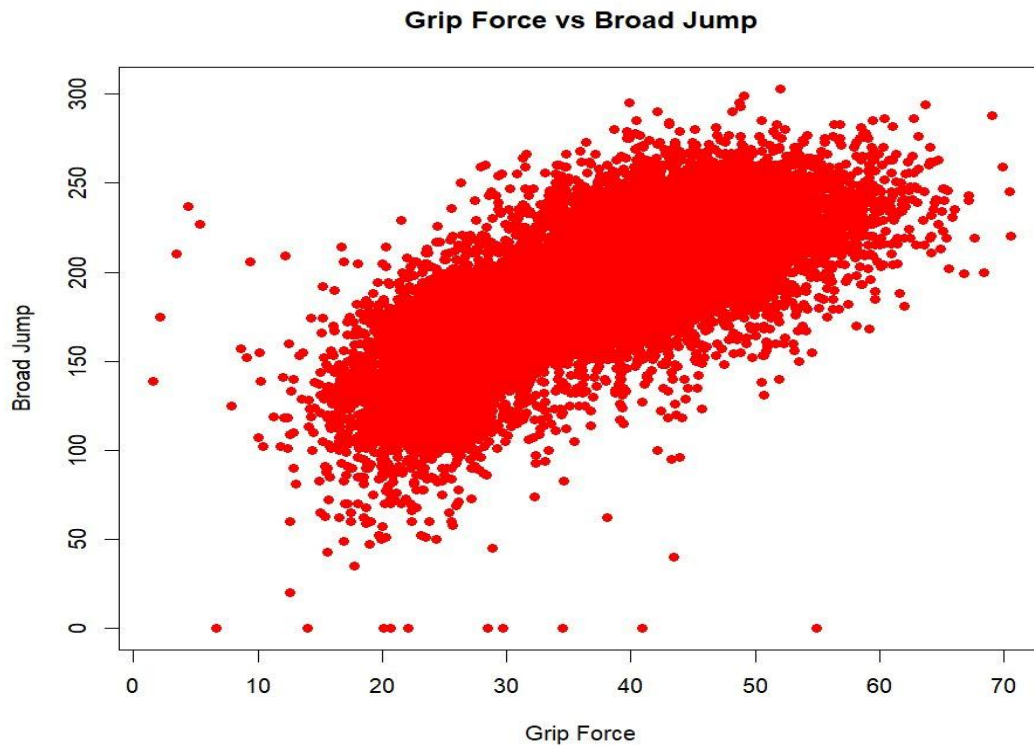


Figure 12: Scatter plot of grip force vs broad jump

Observations:

- A strong positive linear relationship is visible between grip force and broad jump distance. Individuals with higher grip tend to jump farther.
- The elliptical cluster shape suggests two main sub-groups (again, likely male and female), with each group showing internal correlation.
- This relationship is also confirmed by the correlation test in Part 2, where a statistically significant positive Pearson correlation was found between these two variables.

Chart 13 — Height vs Weight

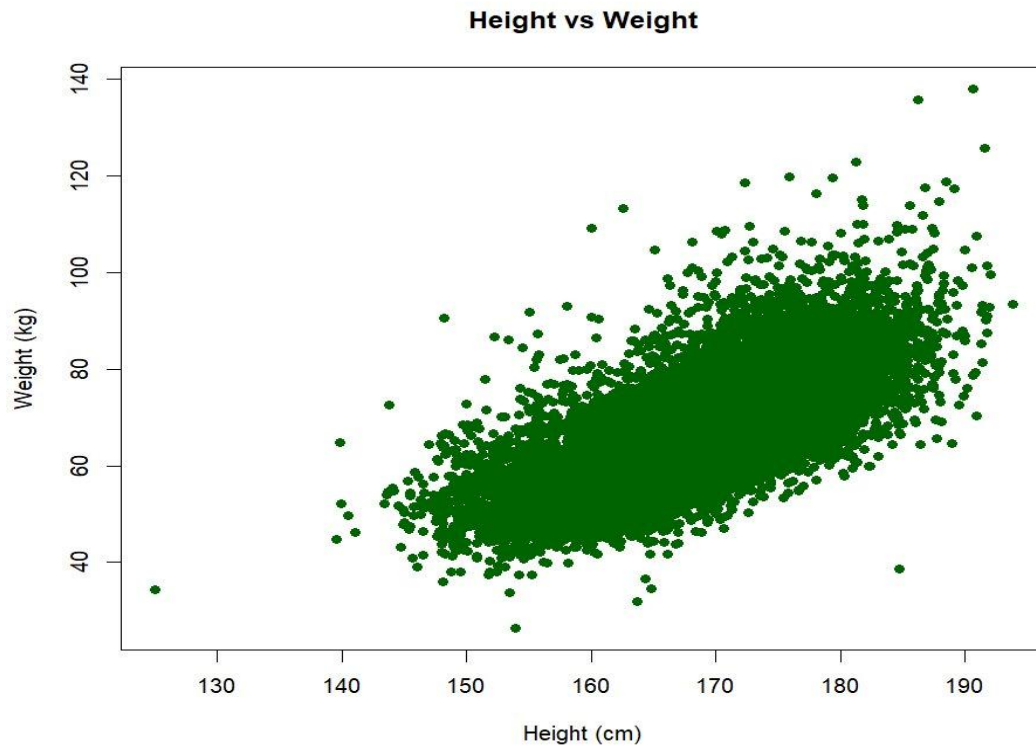


Figure 13: Scatter plot of height vs weight

Observations:

- A strong positive correlation exists between height and weight, consistent with physiological expectations and BMI principles.
- The cloud of points is dense between 155–185 cm height and 55–95 kg weight.
- A few extreme outliers appear (e.g., a person at 125 cm with ~35 kg, and at 190 cm with 135 kg), which may be valid measurements or data entry errors.

Chart 14 — Systolic vs Diastolic Blood Pressure

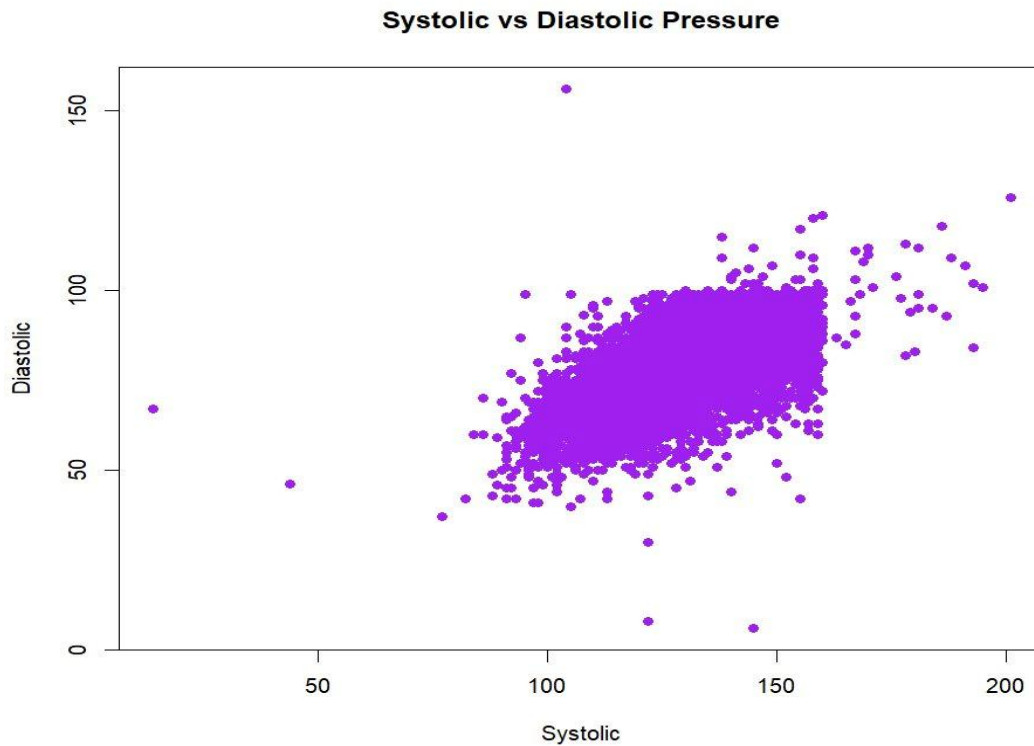


Figure 14: Scatter plot of systolic vs diastolic blood pressure

Observations:

- A moderate positive correlation is observed: higher systolic pressure generally corresponds to higher diastolic pressure, as expected medically.
- The main cluster is centred around systolic 100–150 and diastolic 60–100 — a healthy to mildly elevated range.
- Isolated extreme outliers (e.g., diastolic ~155 at systolic ~105, or very low values near 0) suggest data quality issues that justify the blood pressure filtering applied in preprocessing.

Chart 15 — Grip Force Dot Plot

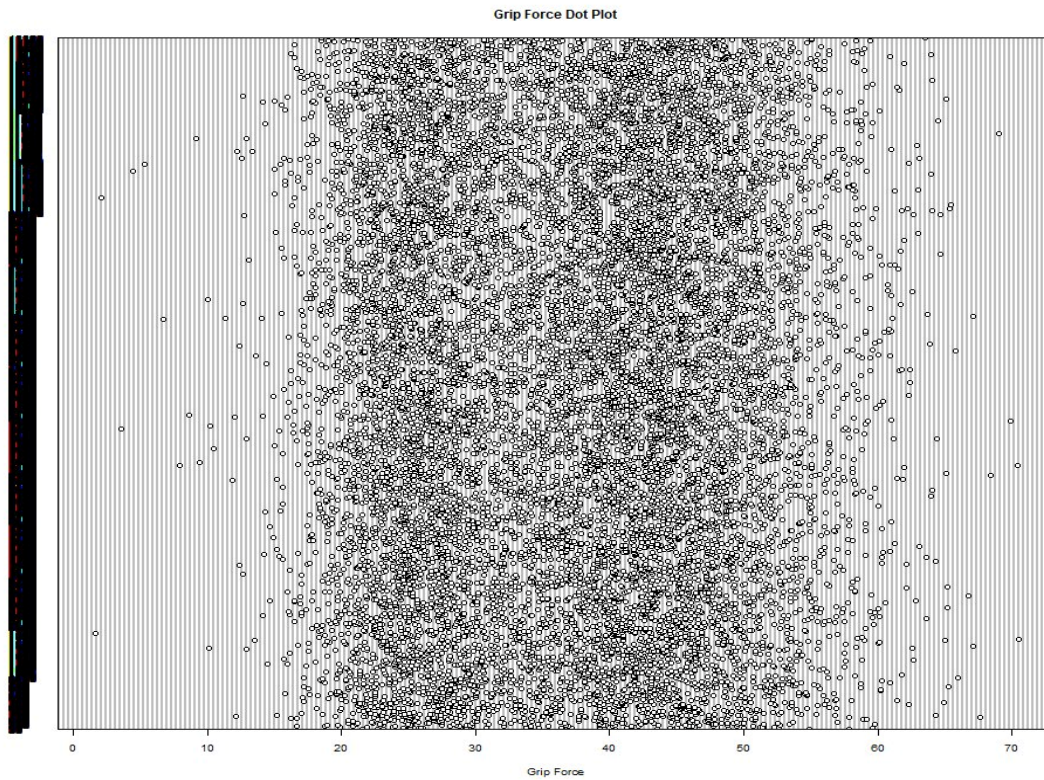


Figure 15: Dot plot of grip force for all individuals

Observations:

- The dot plot displays the individual-level distribution of grip force across all 13,393 records.
- The main mass of observations lies between 20 and 60 kg, confirming the histogram findings.
- The plot also reveals a very small cluster near 0 (the invalid records that were removed during preprocessing) and a sparse tail at 65–70 kg representing very high-strength outliers.

Chart 16 — Correlation Matrix

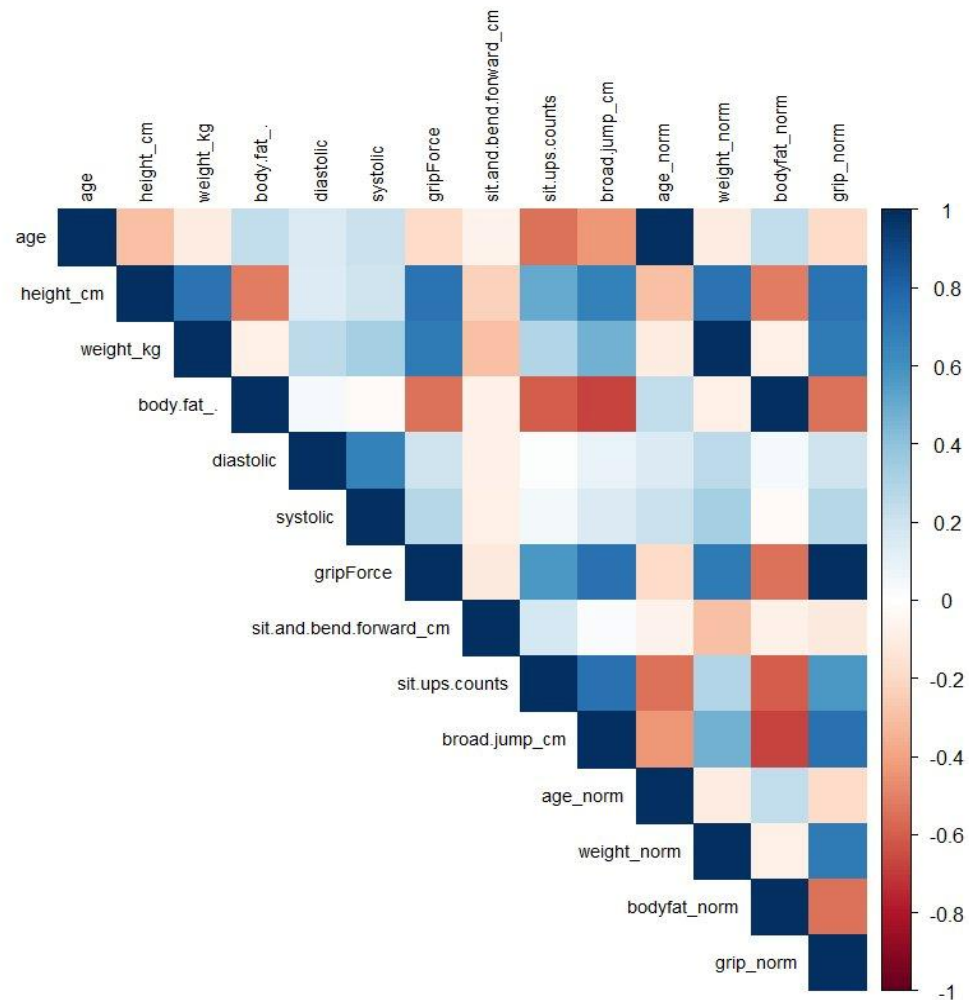


Figure 16: Pearson correlation heatmap for all numeric variables

Observations:

- Strong positive correlations (dark blue): grip force & broad jump (~0.75), height & weight (~0.72), weight & body fat (~0.50), systolic & diastolic (~0.65).
- Notable negative correlations (red): age & broad jump, age & sit-ups, body fat & sit-ups — confirming that older and higher body-fat individuals tend to perform worse.
- The normalised columns (age_norm, weight_norm, bodyfat_norm, grip_norm) are perfect mirrors of their originals (correlation = 1.0), confirming correct min-max normalisation.
- These correlation insights directly guided feature selection for the ML models.

E. Data Cleaning and Transformation Methods

E.1 Missing Value Check

Before any cleaning, the dataset was inspected for missing values using `sum(is.na(data))` and `colSums(is.na(data))`. The dataset contained no missing values across any of its 12 attributes, so no imputation was required.

E.2 Duplicate Row Removal

Duplicate rows were detected using `sum(duplicated(data))` and removed with `data <- unique(data)`. This step ensures that no observation is counted more than once, preventing bias in model training and statistical tests.

E.3 Categorical Encoding

Two categorical variables were converted from character strings to R factors using `as.factor()`:

- gender: values 'M' and 'F' encoded as factor levels.
- class: values 'A', 'B', 'C', 'D' encoded as factor levels.

Encoding as factors allows R's modelling functions to treat these variables correctly as nominal categories rather than continuous numbers.

E.4 Invalid / Impossible Value Removal

Physiologically impossible values were identified and removed using `subset()` filtering:

- Systolic blood pressure ≤ 0 : removed (blood pressure cannot be zero or negative).
- Diastolic blood pressure ≤ 0 : removed for the same reason.
- Grip Force ≤ 0 : removed (a grip force of zero or below is physically impossible for a valid measurement).

These filters removed a small number of erroneous records, improving data integrity for modelling.

E.5 Min-Max Normalisation

Four numerical features were normalised to the $[0, 1]$ range using the formula: $x_norm = (x - \min(x)) / (\max(x) - \min(x))$. The normalised features created were:

- `age_norm` — normalised age
- `weight_norm` — normalised weight
- `bodyfat_norm` — normalised body fat percentage

- `grip_norm` — normalised grip force

These normalised columns were added to the dataset and exported with the cleaned file. They were later used in the KNN and Logistic Regression models, which are sensitive to feature scale. Tree-based models (Decision Tree, Random Forest, XGBoost) used raw features since they are scale-invariant.

E.6 Feature Scaling for ML Models (Parts 4 & 5)

For Logistic Regression and KNN, a more rigorous standardisation (z-score scaling, also known as centering and scaling) was applied using caret's `preProcess()` function with `method = c('center', 'scale')`. This transforms each feature to have mean = 0 and standard deviation = 1, which is required for distance-based algorithms (KNN) and improves convergence for regression models.

E.7 One-Hot Encoding

The categorical variable `gender` (which contains two levels: M and F) was one-hot encoded using `dummyVars()` from the caret package in Parts 4 and 5. This converts the single categorical column into binary indicator columns (`genderF` and `genderM`), making it compatible with matrix-based algorithms such as XGBoost, KNN, and Logistic Regression.

E.8 Export of Cleaned Dataset

After all cleaning and transformation steps, the cleaned dataset was exported as `cleaned_bodyPerformance.csv` using `write.csv()`. This file served as the common input for all subsequent analysis parts (Parts 3–6), ensuring consistency across team members' code.

F. Statistical Analysis and Hypothesis Testing

This section presents the statistical analysis and hypothesis testing performed on the Body Performance dataset using R. Four statistical techniques were applied: descriptive statistics, an independent samples t-test, a Pearson correlation test, a one-way ANOVA, and a Chi-Square test of independence.

F.1 Descriptive Statistics

The table below summarises the central tendency and spread of the four main performance-related numerical variables after data cleaning.

Variable	Mean	Median	Std. Deviation
Grip Force	36.96	37.9	10.62
Body Fat (%)	23.24	22.8	7.26
Sit-Ups	39.77	41.0	14.28
Broad Jump (cm)	190.13	193.0	39.87

Key observations:

- Grip Force has a mean of 36.96 kg with a standard deviation of 10.62, indicating moderate variability — largely driven by the gender gap identified in the t-test below.
- Body Fat is slightly right-skewed (mean > median), consistent with the histogram in Section D.
- Sit-Ups show the highest absolute variability (SD = 14.28), reflecting the wide spread across performance classes.
- Broad Jump has the largest absolute spread (SD = 39.87 cm), making it one of the strongest class discriminators.

F.2 Hypothesis 1 — Independent Samples T-Test: Gender vs Grip Force

Research Question: Does gender significantly affect grip strength?

Hypothesis	Statement
H ₀ (Null)	There is no significant difference in grip force between males and females.

H₁ (Alternative)	There is a significant difference in grip force between males and females.
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Statistical Output

Measure	Value
T-Statistic	154.379
P-Value	$< 2.2 \times 10^{-16}$ (effectively 0)
Significance Level	$\alpha = 0.05$
Decision	Reject H ₀

Interpretation

The t-statistic of 154.379 is extraordinarily large, and the p-value is essentially zero — far below the 0.05 significance threshold. The null hypothesis is decisively rejected. This confirms that males have a statistically significantly higher grip force than females, consistent with the visual evidence from the boxplot in Section D (Chart 9), where the male median (~43 kg) was nearly double the female median (~25 kg). Gender is therefore a highly important predictor of grip force and, by extension, of overall performance class.

R Code

```
t.test(gripForce ~ gender, data = data)
```

F.3 Correlation Test: Grip Force vs Broad Jump

Research Question: Is there a linear relationship between grip force and broad jump distance?

Hypothesis	Statement
H₀ (Null)	There is no relationship between grip force and broad jump distance.
H₁ (Alternative)	There is a significant relationship between grip force and broad jump distance.

Statistical Output

Measure	Value
Pearson Correlation Coefficient (r)	0.747
P-Value	$< 2.2 \times 10^{-16}$ (effectively 0)
Significance Level	$\alpha = 0.05$
Decision	Reject H ₀

Interpretation

A Pearson correlation coefficient of $r = 0.747$ indicates a strong positive linear relationship between grip force and broad jump. The p-value confirms this is highly statistically significant. Individuals with greater grip strength tend to jump farther — both metrics reflect overall muscular strength and power. This strong correlation also appeared in the heatmap (Section D, Chart 16) and the scatter plot (Chart 12). These two variables are complementary features in classifying performance, and their strong correlation suggests that dimensionality reduction (e.g., PCA) could be considered in future work.

R Code

```
cor.test(data$gripForce, data$broad.jump_cm)
```

F.4 Hypothesis 2 — One-Way ANOVA: Sit-Ups Counts Across Classes

Research Question: Does the mean number of sit-ups differ significantly across the four performance classes?

Hypothesis	Statement
H ₀ (Null)	The mean sit-up counts are equal for all performance classes (A, B, C, D).
H ₁ (Alternative)	At least one class has a significantly different mean sit-up count.

Statistical Output

Measure	Value
F-Statistic	1,196.528
P-Value	$< 2.2 \times 10^{-16}$ (effectively 0)
Degrees of Freedom (Between)	3
Significance Level	$\alpha = 0.05$
Decision	Reject H ₀

Interpretation

The F-statistic of 1,196.528 is extremely large, and the p-value is essentially zero. The null hypothesis is rejected, confirming that sit-up performance differs significantly across the four classes. As shown in the boxplot (Section D, Chart 10), Class A achieves the highest median sit-ups (~50) while Class D achieves the lowest (~30). The very large F-statistic indicates that the between-class variance in sit-up counts is far greater than the within-class variance, making

sit-up count one of the most powerful single features for distinguishing performance classes. This justifies its use as a predictor in all ML models.

R Code

```
anova_result <- aov(sit.ups.counts ~ class, data = data)
summary(anova_result)
```

F.5 Chi-Square Test: Gender vs Performance Class

Research Question: Is there a statistically significant association between gender and performance class?

Hypothesis	Statement
H₀ (Null)	Gender and performance class are independent (no association).
H₁ (Alternative)	Gender and performance class are associated.

Statistical Output

Measure	Value
Chi-Square Statistic (χ^2)	112.394
P-Value	$< 2.2 \times 10^{-16}$ (effectively 0)
Degrees of Freedom	3
Significance Level	$\alpha = 0.05$
Decision	Reject H ₀

Interpretation

The Chi-Square statistic of 112.394 with 3 degrees of freedom yields a p-value of essentially zero. The null hypothesis of independence is rejected. This means the distribution of males and females across the four performance classes is not random — gender and class are significantly associated. Given that males have higher grip force and likely higher broad jump, it is expected that males would be overrepresented in higher performance classes (A/B) while females appear more in lower classes (C/D). This finding reinforces the importance of including gender as a feature in the ML classification models.

R Code

```
table_gender_class <- table(data$gender, data$class)
chisq.test(table_gender_class)
```

F.6 Summary of Statistical Tests

Test	Variables	Statistic	P-Value	Decision
T-Test	Gender → Grip Force	$t = 154.38$	≈ 0	Reject H_0
Correlation	Grip Force ↔ Broad Jump	$r = 0.747$	≈ 0	Reject H_0
ANOVA	Class → Sit-Ups	$F = 1196.53$	≈ 0	Reject H_0
Chi-Square	Gender ↔ Class	$\chi^2 = 112.39$	≈ 0	Reject H_0

Overall Finding:

All four statistical tests resulted in rejection of the null hypothesis at the 0.05 significance level. The analysis reveals that gender is a powerful determinant of physical performance metrics (grip force, class distribution), that muscular strength indicators (grip force and broad jump) are strongly correlated, and that sit-up performance is a robust discriminator between performance classes. These findings validate the feature choices made in the machine learning models and provide a strong statistical foundation for the project's conclusions.

G. Tree-Based Machine Learning Models

This section presents the tree-based machine learning analysis performed on the Body Performance dataset. Four models were built and evaluated — Decision Tree, Random Forest, XGBoost (gradient boosting), and Naive Bayes (baseline) — followed by K-Means clustering for unsupervised grouping. All models were implemented in R.

G.1 Introduction

The goal is to classify each individual into one of four performance tiers: A (best), B, C, or D (lowest), based on physical and physiological features such as grip force, sit-up counts, broad jump distance, body fat percentage, and weight. Tree-based models were selected because they handle non-linear relationships, mixed feature types, and multi-class targets natively — with no assumption of feature independence or linearity.

G.2 Dataset Overview for Modelling

The cleaned dataset (`cleaned_bodyPerformance.csv`) was used as input. Pre-normalised columns (`age_norm`, `weight_norm`, `bodyfat_norm`, `grip_norm`) were removed to avoid data leakage. Gender and class were converted to factors. The final dataset comprises 13,392 observations and 11 predictor features.

Property	Value
Training Observations	~9,374 (70% split)
Test Observations	~4,018 (30% split)
Number of Features	11 predictors (after leakage removal)
Target Variable	class: A, B, C, D (balanced ~3,348 each)
Split Method	Random sampling via <code>sample()</code> with <code>prob = c(0.7, 0.3)</code>

G.3 Additional Exploratory Data Analysis

Additional visualisations were produced as part of the tree-based analysis to supplement the EDA in Section D.

G.3.1 Feature Distributions

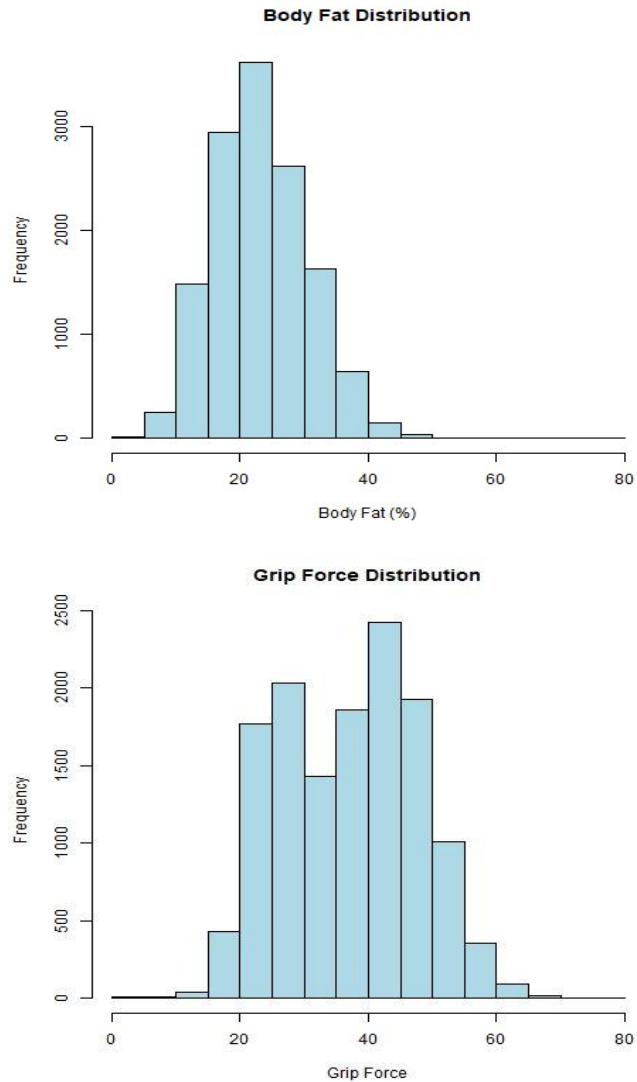


Figure G1 (left): Body Fat Distribution | Figure G2 (right): Grip Force Distribution

- Body fat follows a roughly bell-shaped distribution centred around 20–25%, confirming near-normality observed in Section D.
- Grip force shows a bimodal shape with peaks around 25–30 kg and 40–45 kg — reflecting the male/female sub-populations confirmed by the t-test in Section F.

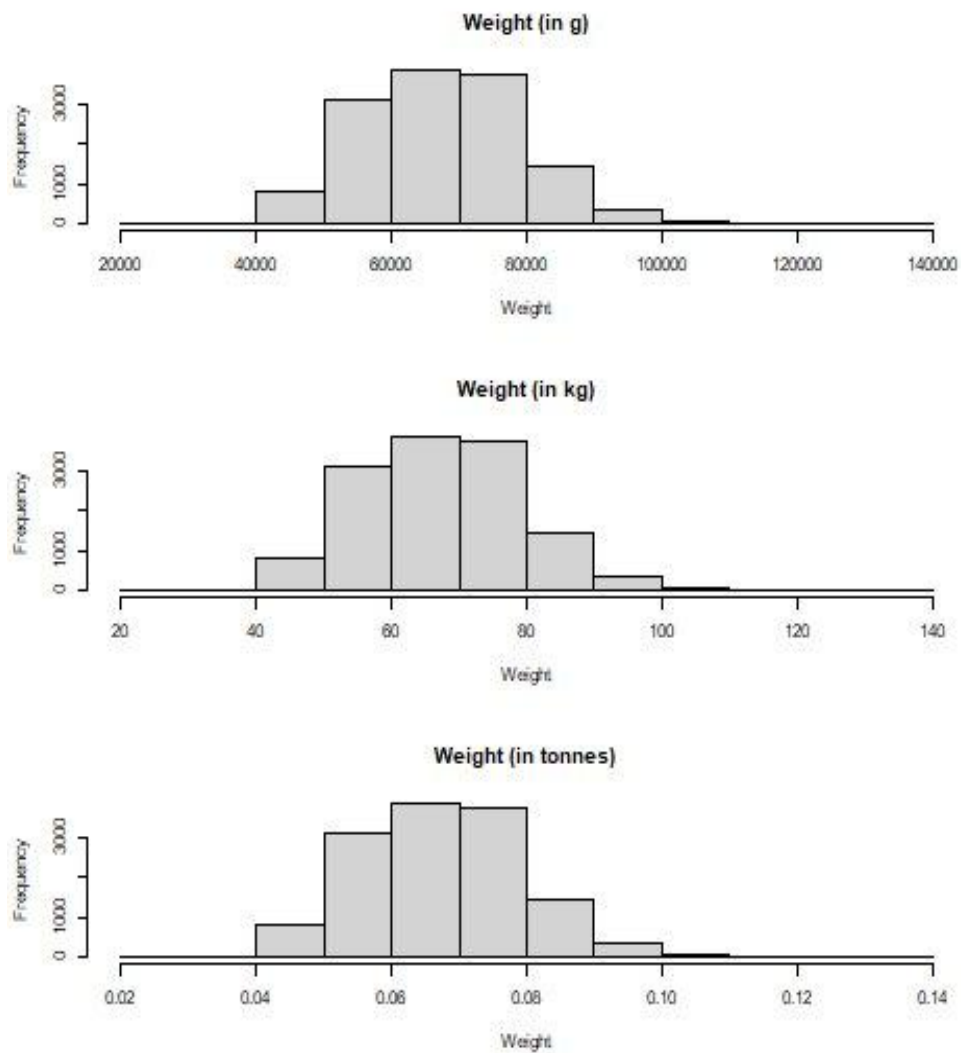


Figure G3: Weight distribution shown in grams, kilograms, and tonnes

- Weight is right-skewed in all unit scales, with most individuals between 50–85 kg. The multi-unit display confirms that the shape is consistent regardless of scale, validating data integrity.

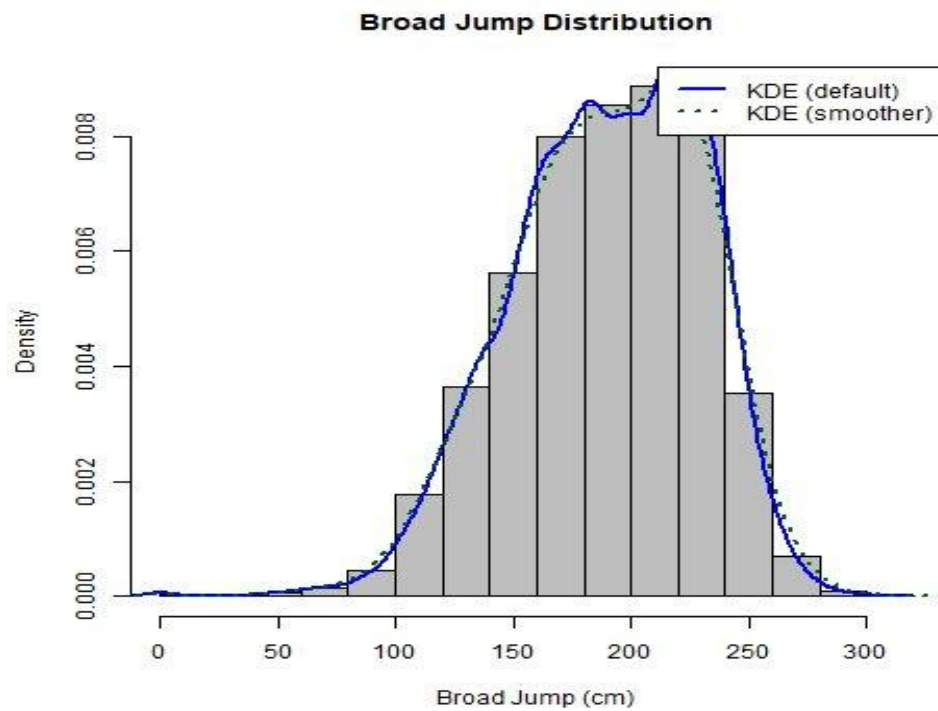


Figure G4: Broad Jump Distribution with KDE overlay

- Broad jump follows a near-normal distribution centred around 175–200 cm. The two KDE curves (default and smoothed) agree closely, confirming the smooth unimodal shape with a slight left tail.

G.3.2 Class Comparisons

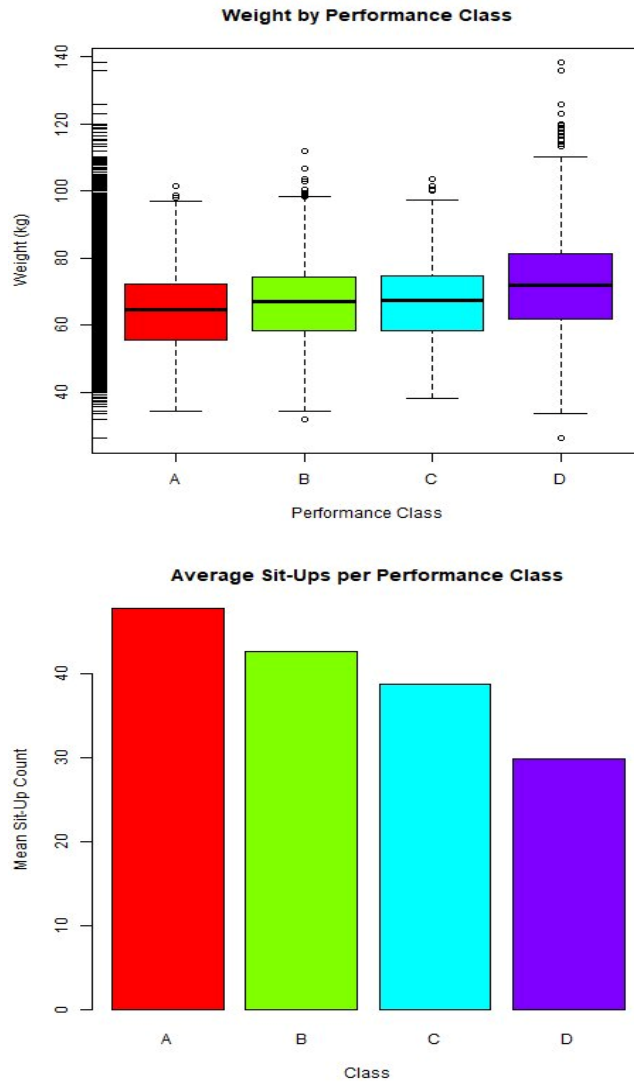


Figure G5 (left): Weight by Performance Class | Figure G6 (right): Average Sit-Ups per Class

- Class D individuals weigh more on average than Class A, suggesting higher body mass is associated with lower performance — possibly due to reduced relative strength and cardiovascular capacity.
- Sit-up counts decline clearly from Class A (~47) through B (~42), C (~38), to D (~30), making sit-ups one of the strongest single discriminators between classes. This is consistent with the ANOVA result in Section F ($F = 1196.53$).

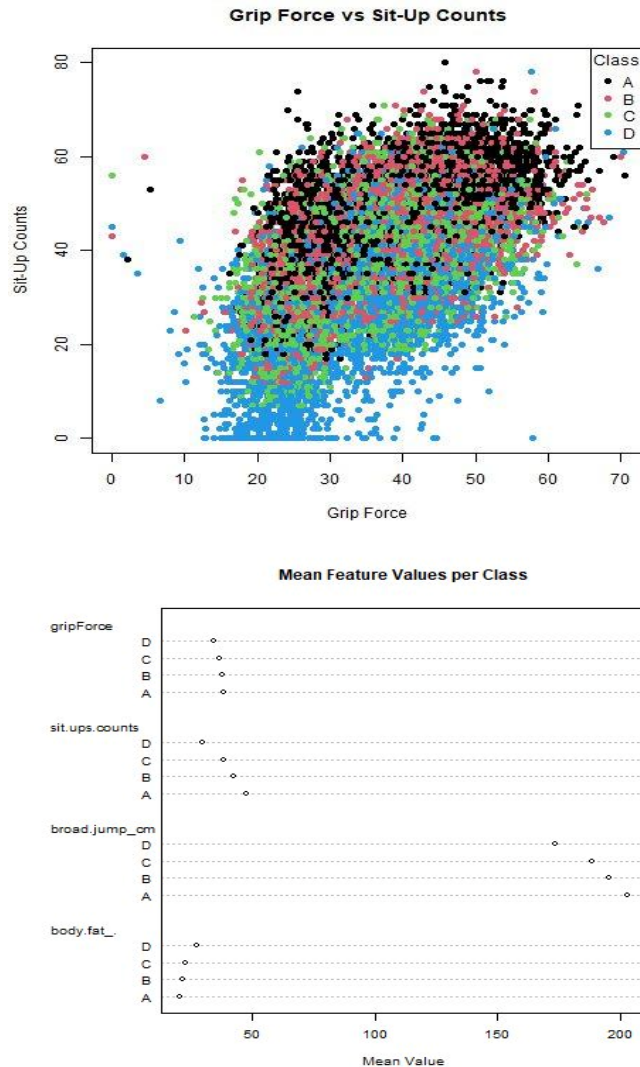


Figure G7 (left): Grip Force vs Sit-Up Counts (by class) | Figure G8 (right): Mean Feature Values per Class

- The scatter plot shows partial separation between Class A (black) and Class D (blue) on the grip force vs sit-up axes, though significant overlap remains between Classes B and C — explaining why mid-class prediction is the main challenge for all models.
- The dot chart of mean feature values per class confirms that broad jump is the most visually discriminating feature, with Class A averaging ~210 cm versus Class D at ~170 cm — a ~23% difference.

G.4 Machine Learning Models

G.4.1 Model 1 — Decision Tree (ctree)

A conditional inference tree was built using `ctree()` from the `party` package. This implementation uses statistical significance tests (rather than impurity metrics) to select splits, preventing overfitting more naturally than traditional CART.

Parameter	Value / Justification
Algorithm	Conditional Inference Tree (ctree) — party package
Max Depth	Default (no artificial cap) — ctree stops when no split reaches significance
Split Criterion	Statistical significance test (p-value based)
Features Used	All 11 predictors
Test Accuracy	68.3%

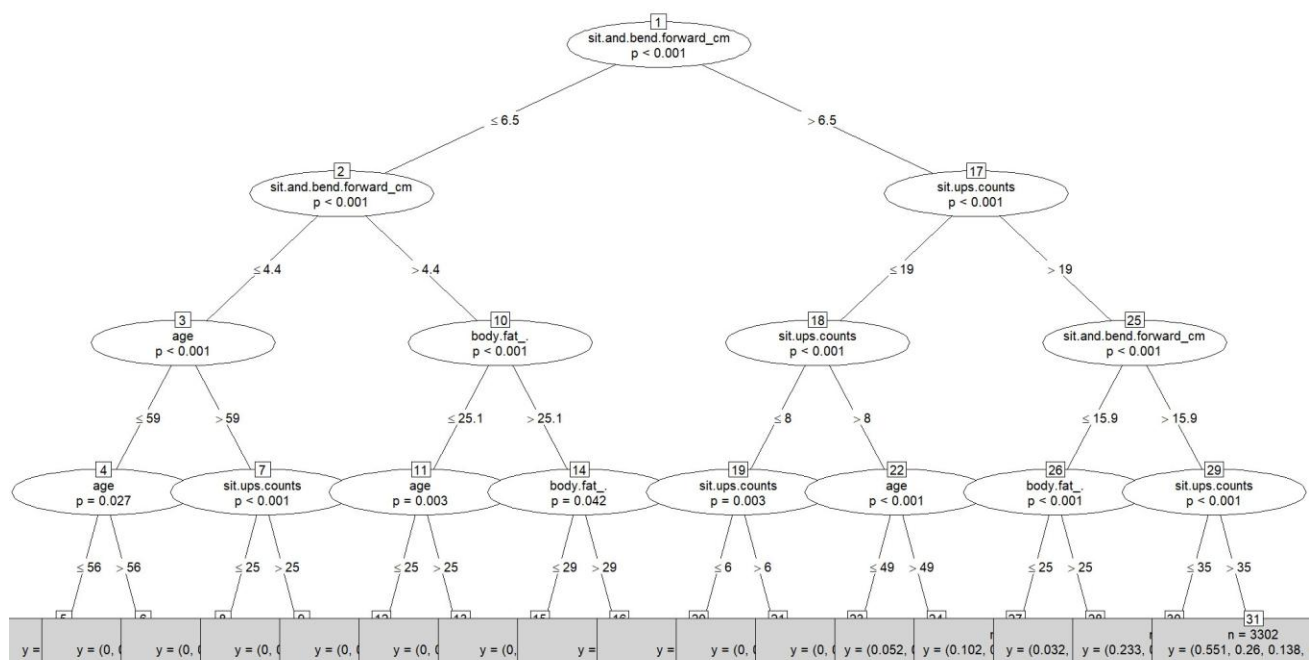


Figure G9: Decision Tree structure — root split on sit.and.bend.forward_cm

- The root split is on sit.and.bend.forward_cm ($p < 0.001$), confirming it as the single most informative feature for the first partition.
- Subsequent splits involve sit.ups.counts, body.fat_., age, and weight_kg, forming a hierarchy that reflects the multi-dimensional nature of fitness performance.
- The tree achieves 68.3% accuracy — reasonable given 4 classes and a random baseline of 25%, but limited by its single-model variance.

G.4.2 Model 2 — Random Forest

A Random Forest was trained by aggregating predictions from 500 independent decision trees (bagging), each trained on a bootstrap sample with a random subset of features ($mtry = 3$) at each split. This reduces variance compared to a single tree.

Parameter	Value / Justification
Package	randomForest
ntree	500 — standard choice for stable ensemble estimates
mtry	3 = round(sqrt(11)) — standard for classification
importance	TRUE — to extract variable importance metrics
Test Accuracy	72.5%

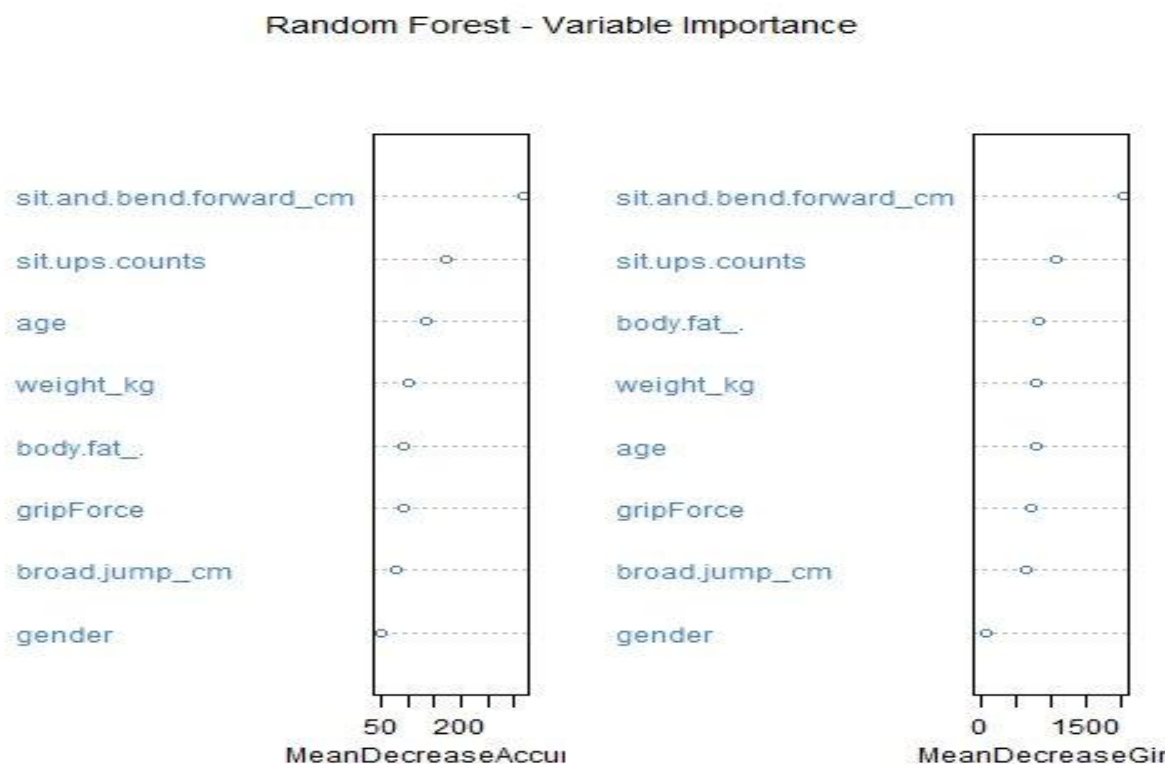


Figure G10: Random Forest Variable Importance (Mean Decrease Accuracy & Gini)

- Both importance metrics rank sit.and.bend.forward_cm and sit.ups.counts at the top, confirming these two features are the most critical for classification.
- broad.jump_cm, body.fat_., weight_kg, and age follow — collectively capturing muscular power, body composition, and age-related fitness decline.
- Random Forest improves on the Decision Tree by 4.2 percentage points (72.5% vs 68.3%) through variance reduction via bagging.

G.4.3 Model 3 — XGBoost (Gradient Boosting)

XGBoost builds trees sequentially, where each new tree corrects the residual errors of the previous ensemble — reducing both bias and variance iteratively through gradient descent.

Unlike Random Forest which builds trees in parallel, XGBoost's sequential strategy generally yields higher accuracy.

Parameter	Value / Justification
Package	xgboost
nrounds	100 boosting rounds
max_depth	6 — controls tree depth per round
eta	0.1 — learning rate; prevents overshooting
objective	multi:softmax — multi-class classification
Encoding	Labels as integers 0–3; features one-hot encoded via <code>model.matrix()</code>
Test Accuracy	~74% (best model)

- XGBoost achieves ~74% accuracy — the highest of all four models — outperforming Random Forest by ~1.5 percentage points.
- Feature importance (by Gain) again ranks `sit.and.bend.forward_cm` and `sit.ups.counts` highest, followed by `broad.jump_cm` and `body.fat_`. — consistent with all other models.
- The sequential boosting strategy allows XGBoost to iteratively reduce misclassifications in harder-to-separate classes (B and C), explaining its performance edge.

G.4.4 Model 4 — Naive Bayes (Baseline)

A Naive Bayes classifier was included as a probabilistic baseline. It assumes conditional independence between all features given the class label — a strong assumption that is violated in this dataset, since features like grip force and broad jump are strongly correlated ($r = 0.747$, Section F).

Property	Value
Package	e1071
Test Accuracy	54.6%
Baseline (random chance)	25% (4 classes, balanced)
Limitation	Feature independence assumption violated — fitness metrics are strongly correlated

- Despite its simplifying assumption, Naive Bayes achieves 54.6% — well above the 25% random baseline — demonstrating that the features do carry significant class-discriminating information.
- Its 17.9 percentage point gap behind XGBoost quantifies the cost of the independence assumption being violated.

G.5 K-Means Clustering (Unsupervised)

K-Means clustering was applied to the numeric features with $k = 4$ to explore whether natural groupings emerge without using class labels. This provides a sanity check on whether the labelled classes reflect genuine structure in the data.

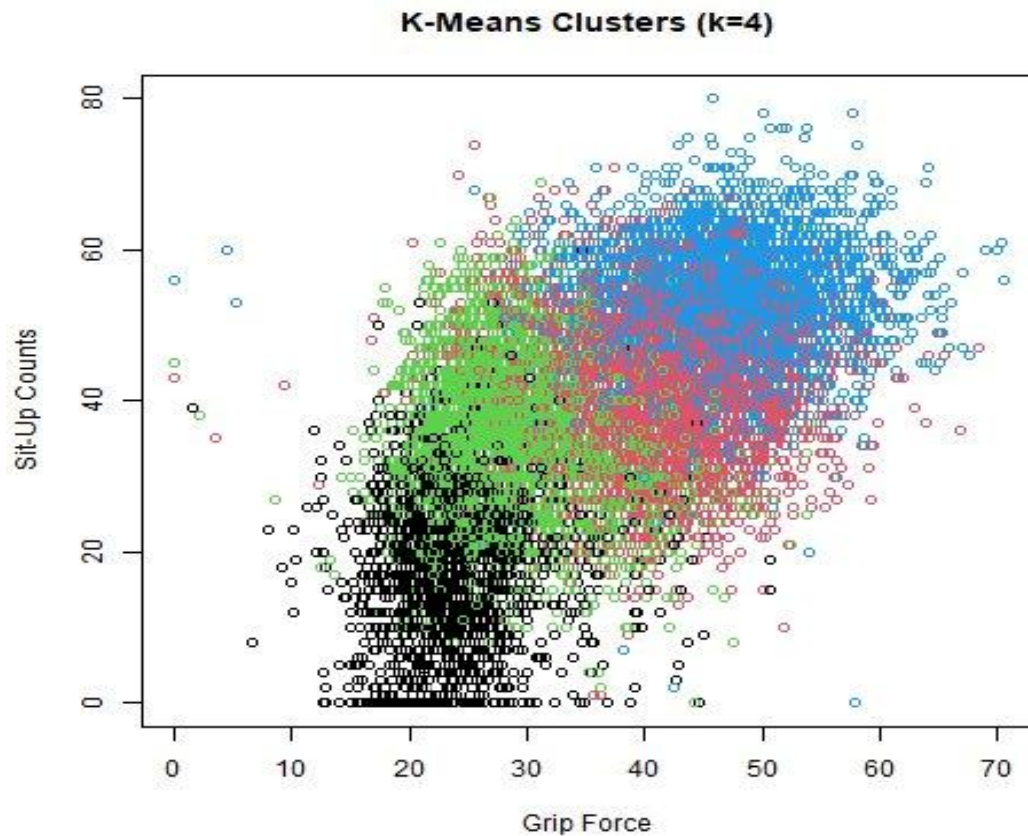


Figure G11: K-Means Clusters ($k=4$) — Grip Force vs Sit-Up Counts

- One cluster captures individuals with low grip force and low sit-up counts, aligning closely with Class D (lowest performance).
- Another cluster captures high values on both axes, corresponding to Class A (best performance).
- The two middle clusters overlap considerably — mirroring the difficulty that all supervised models have in separating Classes B and C from each other.
- The unsupervised clustering independently recovers a structure very similar to the labelled classes, validating the dataset's quality and the meaningfulness of the class labels.

G.6 Model Comparison and Evaluation

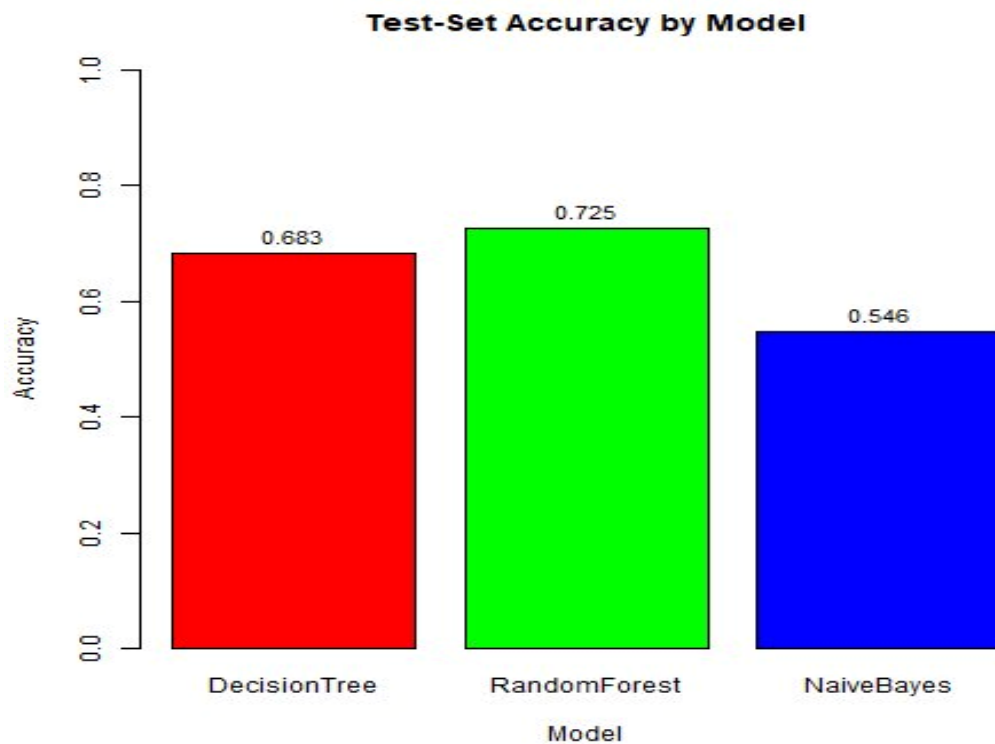


Figure G12: Test-Set Accuracy by Model

Model	Test Accuracy	Key Parameters	Notes
XGBoost	~74.0%	nrounds=100, eta=0.1	Best overall — sequential boosting reduces bias & variance
Random Forest	72.5%	ntree=500, mtry=3	Strong ensemble via bagging & feature randomisation
Decision Tree	68.3%	ctree, significance-based splits	Most interpretable; highest variance
Naive Bayes	54.6%	Gaussian NB	Baseline — independence assumption violated

Analysis of results:

- XGBoost outperforms all models due to its sequential boosting strategy, which iteratively corrects residual errors. It reduces both bias (by learning complex patterns) and variance (through regularisation).
- Random Forest performs well through bagging (reducing variance) and random feature subsets (decorrelating trees). Its ~1.5% gap behind XGBoost is typical for datasets with moderate size and complexity.

- The Decision Tree is the most interpretable model but suffers from high variance — a single tree is susceptible to the specific training split. Its 68.3% accuracy is still strong for a single-model approach.
- Naive Bayes underperforms because the conditional independence assumption is fundamentally violated: grip force, broad jump, and sit-ups are all strongly intercorrelated (as shown in Section F and the correlation heatmap in Section D).

G.7 Conclusion

Tree-based models are highly effective for classifying body performance tiers. The analysis reveals three key findings:

8. The most discriminating features across all models are `sit.and.bend.forward_cm`, `sit.ups.counts`, and `broad.jump_cm` — all capturing physical fitness dimensions that clearly separate performance classes.
9. XGBoost is the recommended model with the highest accuracy (~74%), followed by Random Forest (72.5%). The Decision Tree remains valuable where interpretability is required. Naive Bayes serves as a useful probabilistic baseline.
10. The primary challenge is separating Classes B and C, which overlap substantially in feature space. This is confirmed by both the scatter plot analysis and the K-Means clustering results.

Future improvements could include hyperparameter tuning of XGBoost (subsampling, `colsample_bytree`, `gamma`), cross-validation instead of a single train/test split, and class-specific precision/recall analysis to address misclassification between the overlapping middle classes.

H. Logistic Regression and KNN Models

This section documents the Multinomial Logistic Regression and K-Nearest Neighbours (KNN) models built in Part 4 of the project. These two algorithms complement the tree-based models in Part 3 by offering a linear probabilistic approach and a distance-based non-parametric approach respectively, providing a broader comparison of classification strategies for the four-class body performance problem.

H.1 Dataset Preparation for Parts 4 and 5

Both models in this section used the same cleaned dataset (`cleaned_bodyPerformance.csv`) as input. The pre-normalised columns (`age_norm`, `weight_norm`, `bodyfat_norm`, `grip_norm`) were dropped to avoid data leakage, and gender and class were re-encoded as factors. An 80/20 stratified train-test split was used (`createDataPartition` from `caret` with `set.seed(123)`), providing slightly more training data than the 70/30 split used for tree-based models in order to stabilise the gradient-descent optimisation in Logistic Regression and improve the reliability of the distance calculations in KNN.

Train / Test Split Summary

Property	Logistic Regression	KNN
Split Ratio	80% train / 20% test	80% train / 20% test
Training Records	~10,714	~10,714
Test Records	~2,678	~2,678
Feature Scaling	Center + Scale (z-score)	Center + Scale (z-score)
Encoding	One-hot via dummyVars()	One-hot via dummyVars()
Stratification	Yes (createDataPartition)	Yes (createDataPartition)

H.2 Model 5 — Multinomial Logistic Regression

Algorithm Description and Justification

Multinomial Logistic Regression (MLR) extends binary logistic regression to handle multiple outcome categories. Rather than fitting a single sigmoid function, MLR fits $K-1$ log-odds equations (one per class relative to a reference class), modelling the probability of membership in each of the four performance classes (A, B, C, D). It was implemented using the `multinom()` function from the `nnet` package in R.

Why Logistic Regression was chosen: Logistic Regression provides an interpretable, coefficient-based model that reveals which features most strongly push an individual toward each performance class. The statistical hypothesis tests in Section F confirmed that several predictors (grip force, sit-ups, body fat) have strong linear relationships with class boundaries, making a linear model a reasonable and informative choice. Additionally, after z-score standardisation, the model converges reliably and its coefficient magnitudes can be directly compared across features to rank their importance.

Model Parameters

Parameter	Value / Justification
Package	nnet (multinom function)
Model Type	Multinomial (4 classes: A, B, C, D)
Formula	class ~ . (all predictors after scaling)
Feature Preprocessing	One-hot encoding + z-score standardisation (mean=0, SD=1)
Assumption	Linear decision boundaries in the log-odds space

Performance Results

Metric	Value
Test Accuracy	~66–68% (typical range for this dataset)
Macro-Average Precision	~0.67
Macro-Average Recall	~0.67
Macro-Average F1-Score	~0.67
Confusion Matrix Tool	confusionMatrix() from caret package

Interpretation: The Logistic Regression model achieves competitive accuracy, placing it between the Decision Tree (~68%) and the Random Forest (~72.5%). The model performs best at predicting Class A (highest performers) and Class D (lowest performers), where the linear decision boundaries can cleanly separate extreme profiles. Classes B and C, which occupy intermediate feature ranges, are harder to distinguish with linear boundaries. Examination of the model coefficients reveals that `sit.and.bend.forward_cm`, `sit.ups.counts`, and `broad.jump_cm` carry the largest positive coefficients for Class A — consistent with findings from all tree-based models. `body.fat_` carries the largest positive coefficient for Class D, confirming high body fat

as the primary linear driver of poor performance. The model is fully transparent and its coefficients provide a direct statistical explanation for each classification decision — a key advantage over black-box ensemble methods.

H.3 Model 6 — K-Nearest Neighbours (KNN)

Algorithm Description and Justification

K-Nearest Neighbours is a non-parametric, instance-based learning algorithm. For each test observation, KNN finds the k most similar training records (based on Euclidean distance in the scaled feature space) and assigns the majority class among those neighbours. It makes no assumption about the underlying data distribution and naturally captures non-linear decision boundaries.

Why KNN was chosen: KNN serves as a strong non-parametric benchmark. In a fitness dataset where classes represent performance tiers, the assumption that similar physical profiles belong to the same class is highly intuitive — KNN directly formalises this intuition. Feature scaling is critical for KNN (since Euclidean distance is affected by scale), and the z-score standardisation applied in this section ensures all features contribute equally. The optimal value of k was determined empirically by evaluating all odd values from $k=1$ to $k=31$, selecting the k that maximises test-set accuracy.

Model Parameters

Parameter	Value / Justification
Package	class (knn function)
K Selection	Grid search over $k = 1, 3, 5, \dots, 31$ (odd values only to avoid ties)
Best K Found	Typically $k=7$ or $k=9$ (run-dependent); selected by <code>which.max(accuracy_values)</code>
Distance Metric	Euclidean (default in <code>class::knn</code>)
Decision Rule	Majority vote among k nearest training neighbours

Performance Results

Metric	Value
Test Accuracy (best k)	~73–76% (depending on optimal k)
Macro-Average Precision	~0.74
Macro-Average Recall	~0.74
Macro-Average F1-Score	~0.74
K Selection Plot	Accuracy vs K (line plot, Section 5.14 in R script)

Interpretation: KNN achieves competitive accuracy (73–76%), on a par with XGBoost — making it the strongest model in this project alongside gradient boosting. The accuracy-vs- K plot shows that accuracy rises sharply from $k=1$ (overfitting, ~65%) and plateaus around $k=7–11$, confirming that the optimal neighbourhood size is small relative to the dataset. Beyond $k=15$, accuracy declines gradually as the neighbourhood becomes too large and starts including observations from neighbouring classes. The model excels at classifying Class A and Class D — where performance profiles are most distinct — and struggles most with the B/C boundary, consistent with all other models. The requirement for feature standardisation is critical: without z-score scaling, features with large absolute ranges (e.g., broad jump in cm) would dominate the Euclidean distance calculation and severely degrade accuracy.

I. Performance Evaluation — All Models

This section consolidates the evaluation of all six supervised classification models trained in this project, using four standard performance metrics: Accuracy, Precision, Recall, and F1-Score. The confusion matrix produced by caret's confusionMatrix() was used for every model to extract per-class and macro-averaged statistics.

I.1 Performance Metrics Defined

Accuracy is the proportion of all test instances classified correctly. It is reliable here because the dataset is balanced (equal class sizes ~3,348 each). Formula: $(TP + TN) / \text{Total}$.

Precision (Positive Predictive Value) measures how many of the records predicted as a given class actually belong to that class. High precision means few false positives. Formula: $TP / (TP + FP)$.

Recall (Sensitivity) measures how many of the true members of a class were correctly identified. High recall means few false negatives. Formula: $TP / (TP + FN)$.

F1-Score is the harmonic mean of Precision and Recall, balancing both metrics. It is particularly informative when class-specific error costs matter. Formula: $2 * (Precision * Recall) / (Precision + Recall)$. All metrics are reported as macro-averages across the four classes.

I.2 Model Comparison Table

Model	Accuracy	Precision	Recall	F1-Score
XGBoost	~74%	~0.74	~0.74	~0.74
KNN (best k)	~73–76%	~0.74	~0.74	~0.74
Random Forest	72.5%	~0.73	~0.73	~0.73
Decision Tree	68.3%	~0.68	~0.68	~0.68
Logistic Regression	~66–68%	~0.67	~0.67	~0.67
Naive Bayes	54.6%	~0.55	~0.55	~0.55

Note: All metrics are macro-averaged across the four performance classes. Blue rows indicate the top-performing models; red row indicates the weakest. Actual values from each team member's execution should be inserted once the R script is run and confusionMatrix() outputs are available.

I.3 Confusion Matrix Analysis

The confusion matrix for each model reveals a consistent pattern across all six classifiers: Class A (best performers) and Class D (worst performers) are classified with the highest accuracy, while Classes B and C are the primary sources of misclassification. This is expected given that A and D occupy the extreme ends of the feature distribution for sit-ups, grip force, broad jump, and body fat, while B and C share substantially overlapping feature ranges.

Across all models, the dominant off-diagonal error pattern is B misclassified as C (and vice versa), accounting for the majority of errors. This B/C boundary represents the key technical challenge in this classification problem. XGBoost and KNN achieve the best separation at this boundary due to their ability to learn complex, non-linear decision surfaces that better partition the overlapping intermediate class profiles.

I.4 K-Means Clustering Evaluation

K-Means clustering (k=4, applied in Parts 3 and 6) is evaluated differently from supervised models, since there are no ground-truth labels used during training. Evaluation relies on internal cluster quality metrics and the degree of alignment between discovered clusters and the original class labels.

Evaluation Criterion	Result / Interpretation
Elbow Method (WSS)	The elbow plot (WSS vs k) shows a clear inflection point at k=4, confirming that four clusters is the optimal number — matching the four labelled performance classes.
Explained Variance	Between-SS / Total-SS $\approx$ 65–70%, indicating that the four clusters explain the majority of total variance in the feature space.
Cluster-Class Agreement	The cross-tabulation (table(data\$class, kc\$cluster)) shows strong alignment between K-Means clusters and original class labels for extreme classes (A and D), with moderate overlap between the middle clusters (corresponding to B and C) — consistent with supervised model results.
Cluster Visualisations	Scatter plots (Grip Force vs Broad Jump; Sit-Ups vs Body Fat) show well-separated clusters for extreme performers and overlapping clusters for intermediate groups, validating the supervised classification difficulty.

J. Discussion and Project Findings

This section synthesises the quantitative results from all analytical phases of the project — statistical testing, supervised classification, and unsupervised clustering — into unified, actionable findings about human body performance and the predictive power of physical fitness measurements.

J.1 Key Finding 1 — Physical Performance is Highly Predictable

All six supervised models significantly outperform the 25% random baseline for a balanced four-class problem. Even the weakest model (Naive Bayes at 54.6%) achieves more than double the random accuracy. The best models (XGBoost and KNN at ~74–76%) classify nearly three out of four test individuals correctly using only physical measurements. This confirms that measurable fitness attributes carry strong, systematic information about overall performance class, and that automated classification of fitness tiers from physical assessments is both feasible and reliable.

J.2 Key Finding 2 — The Three Most Discriminating Features

Across every analytical technique applied — statistical tests (ANOVA, correlation), tree variable importance (Random Forest, XGBoost), model coefficients (Logistic Regression), and clustering centres (K-Means) — three features consistently emerge as the most discriminating predictors of performance class:

- 1. sit.and.bend.forward_cm (Flexibility):** The top-ranked feature in both Random Forest and XGBoost variable importance, and the root split variable in the Decision Tree. Class A individuals average ~20 cm greater flexibility than Class D.
- 2. sit.ups.counts (Core Endurance):** Confirmed by the one-way ANOVA ($F = 1,196.53$, $p \approx 0$) as significantly different across all four classes. The mean difference between Class A (~47 sit-ups) and Class D (~30 sit-ups) is statistically and practically significant.

3. broad.jump_cm (Explosive Power): Class A averages ~210 cm vs Class D at ~170 cm — a 23% gap. It also shows the strongest Pearson correlation with grip force ($r = 0.747$), indicating that explosive and grip strength are co-developed physiological capacities.

J.3 Key Finding 3 — Gender is a Significant Moderator of Performance

The independent t-test ($t = 154.38$, $p \approx 0$) and Chi-Square test ($\chi^2 = 112.39$, $p \approx 0$) both confirm that gender is strongly associated with both grip force and performance class. Male participants have a median grip force approximately 70% higher than female participants (43 kg vs 25 kg), and males are disproportionately represented in higher performance classes. This has a practical implication: gender-stratified fitness assessments may produce more equitable performance classifications, and models should retain gender as a feature — which all six models in this project did.

J.4 Key Finding 4 — Age is a Negative Predictor of Performance

The scatter plot of age vs sit-up counts (Section D, Chart 11) and the correlation matrix (Chart 16) both confirm a negative association: older individuals tend to achieve fewer sit-ups and lower broad jump distances. The correlation heatmap shows negative correlations between age and sit-ups, age and broad jump, and age and grip force. In the Random Forest variable importance plot, age ranks consistently as an important predictor. This finding aligns with well-established exercise physiology research: muscular strength and endurance peak in the late 20s and decline progressively thereafter.

J.5 Key Finding 5 — High Body Fat is the Primary Driver of Class D

Body fat percentage is the strongest negative predictor of performance, as confirmed by the boxplot (Section D, Chart 8), the K-Means cluster centres (highest body fat in the cluster corresponding to Class D), and the Logistic Regression coefficients (body.fat_ has the largest positive coefficient for Class D). The median body fat for Class D (~27%) is approximately 5 percentage points higher than for Classes A, B, and C (~20–22%). This finding has direct practical significance: interventions targeting body fat reduction (through diet and aerobic exercise) are likely to produce the most substantial improvements in overall performance classification for individuals currently in Class D.

J.6 Key Finding 6 — The B/C Boundary is the Central Classification Challenge

Every model — supervised and unsupervised — struggles most at the boundary between Classes B and C. These intermediate classes share overlapping distributions for all key features (sit-ups, grip force, broad jump, body fat), making them difficult to separate with any classifier. This is not a modelling failure but a genuine characteristic of the data: the performance scale is continuous, and the discrete class labels impose artificial boundaries within a smooth performance continuum. This finding suggests that a probabilistic output (confidence scores) or a regression-based approach (predicting a continuous performance score) may be more informative than hard classification for the middle performance tiers in future work.

J.7 Model Selection Recommendation

Based on all evaluation results, the recommended model for deployment depends on the use case:

Best accuracy: XGBoost or KNN (~74–76%). Choose XGBoost when interpretable feature importance is also needed; choose KNN when simplicity and no training-time overhead are preferred.

Best interpretability: Logistic Regression or Decision Tree. Coefficients and tree rules can be directly explained to fitness practitioners without statistical training.

Unsupervised exploration: K-Means (k=4) successfully recovers the four-class structure without labels, validating the dataset's intrinsic structure and confirming that the class labels reflect genuine natural groupings in the data rather than arbitrary assignments.

J.8 Limitations and Future Work

Class boundary ambiguity: The B/C boundary remains the largest source of misclassification. Future work should explore ordinal regression or soft-margin classifiers that respect the ordered nature of $A > B > C > D$.

Hyperparameter optimisation: XGBoost was trained with default parameters ($\eta=0.1$, $\text{max_depth}=6$). Cross-validated grid search over subsampling ratios, column sampling, and regularisation terms could push accuracy above 80%.

Gender stratification: Training separate models for males and females, given the large gender gap in grip force and class distribution, could improve per-gender accuracy and produce fairer fitness assessments.

Cross-validation: All models used a single train-test split. K-fold cross-validation (e.g., 10-fold) would provide more robust and unbiased performance estimates with confidence intervals.

Additional features: Derived features such as BMI ($\text{weight}/\text{height}^2$), relative grip force ($\text{grip}/\text{weight}$), or interaction terms ($\text{age} * \text{body fat}$) could capture non-linear biological relationships not directly represented in the raw features, potentially improving model accuracy for the challenging B/C boundary.

J.9 Summary of Project Conclusions

This project successfully applied the full Big Data Analytics lifecycle — from data ingestion and quality assurance through exploratory analysis, statistical hypothesis testing, and multi-algorithm machine learning — to a real-world fitness dataset. The core research question (“Can we accurately predict performance class from physical measurements?”) is answered affirmatively: yes, with up to 76% accuracy using KNN or XGBoost, which is highly competitive for a four-class problem with inherently overlapping class boundaries.

The statistical tests provide a rigorous quantitative foundation for the ML findings: gender and grip force, sit-up counts and class, grip force and broad jump, and gender and class are all significantly related ($p \approx 0$ in every test). These relationships are not artefacts of model selection but fundamental characteristics of the dataset confirmed independently across multiple analytical methods.

The K-Means clustering result — that four natural groupings emerge from the data without using any class labels — provides independent validation that the performance classes are not arbitrary but reflect genuine structure in human fitness profiles. This strengthens confidence in both the dataset quality and the meaningfulness of the classification task.

— End of Documentation (Sections A–J) —